MULTIDISCIPLINARY SCIENCE JOURNAL

RESEARCH ARTICLE
Published Online: July 9, 2025
https://doi.org/10.31893/multiscience.2026061

A hermeneutical analysis of human rights protections in the deployment and regulation of AI systems in the Western Balkans

Pjereta Agalliu[a] | Anjeza Liçenji[a]

[a]University of Tirana, Faculty of Law, Albania.

Abstract This study uses a hermeneutical approach to analyze human rights protections in AI regulation across the Western Balkans. Six countries, Serbia, Bosnia and Herzegovina, North Macedonia, Albania, Montenegro, and Kosovo, face challenges in balancing technology and fundamental rights. Through an interpretative analysis of legal texts and policies, this research investigates the understanding and implementation of human rights principles in AI governance. The findings show significant disparities in regulatory approaches, with many countries lacking comprehensive AI legislation despite increased technology use. The hermeneutical lens reveals how historical legacies and EU accession influence interpretations of privacy, nondiscrimination, and transparency. The study identifies gaps between international human rights standards and regional practices, suggesting opportunities for rights-centered AI governance. It concludes with policy recommendations for regional cooperation, capacity building, and contextually appropriate regulations to protect human rights and encourage innovation.

Keywords: artificial intelligence, human rights, Western Balkans, hermeneutics, legal interpretation, regulatory frameworks

1. Introduction

The rapid advancement of artificial intelligence (AI) globally has significant implications for human rights and governance, particularly in the Western Balkans, where it intersects with complex political and societal changes. As these countries strive for European Union integration, regulating AI presents unique challenges and opportunities for human rights protection. The Western Balkans' postsocialist transition and varying institutional capacities necessitate a nuanced understanding of human rights in the context of AI. AI is increasingly integrated into public administration, security, healthcare, and financial services, such as facial recognition in Serbia's "safe city" initiatives and algorithmic decision-making in Kosovo's social welfare (Borici, 2025; ITS, 2024; Krivokapić et al., 2023). Deployments of AI raise critical human rights issues, including privacy and surveillance, necessitating regulatory attention. The Western Balkans' regulatory framework remains fragmented, with countries such as Serbia beginning to develop national AI strategies, whereas others lack dedicated policies (Schoenherr, 2024; ITS, 2024). This inconsistency poses challenges for human rights protection. A hermeneutical approach can analyze how human rights are conceptualized in AI governance, considering cultural and historical contexts. This research explores key questions about human rights interpretations in AI frameworks, cultural influences, and gaps between international standards and regional practices, aiming to enhance the understanding of AI governance and improve human rights protections in the region. This research explores policy and regulatory approaches in the Western Balkans by aligning AI governance with EU standards and human rights principles during their integration. It aims to guide the development of frameworks that protect human rights while fostering technological innovation. The research explores how human rights principles are understood and applied in AI governance in the Western Balkans, considering historical, cultural, and political influences. The study hypothesizes that the interpretation and implementation of human rights in AI governance in the Western Balkans are heavily influenced by each country's historical experiences, such as socialist legacies and postconflict transitions. Current political priorities, such as security and economic development, and EU integration status lead to fragmented approaches that often favor security over human rights. This results in gaps between legal commitments and practical implementation. The hypothesis will be tested through hermeneutical analysis of legal texts and policy documents. The research framework involves both broad analytical goals and specific investigative questions that direct our examination of legal texts, policy documents, and implementation practices. The primary research objective is to analyze how human rights principles are interpreted and implemented in AI governance across six Western Balkan countries, considering historical, cultural, and political influences. The specific goals include examining the formal recognition of rights such as privacy, nondiscrimination, transparency, freedom of expression, and human dignity in AI legislation in Serbia, Bosnia and Herzegovina, North Macedonia, Albania, Montenegro, and Kosovo. It investigates the impact of historical legacies, cultural contexts, and political priorities on human rights in AI governance, identifying patterns of convergence and divergence among countries. The research assesses alignment with international standards, such as the EU AI Act and GDPR. It includes case studies on AI applications, evaluates protective mechanisms, and provides policy recommendations for strengthening human rights in AI governance. This comprehensive analysis contributes to the growing body of scholarship on AI governance and human rights while providing practical insights for policymakers, legal practitioners, and civil society organizations working to strengthen human rights protections in technological contexts. The paper proceeds with a literature review examining human rights principles relevant to AI systems, global AI regulation frameworks, hermeneutical theory in legal analysis, and regional legal frameworks in the Western Balkans. The methodology section details our hermeneutical approach, data sources, and analytical framework. The results and discussion section presents our comparative analysis of AI-related legislation, hermeneutical interpretation of legal texts and policies, case studies of AI deployment, and identification of challenges and opportunities. The paper concludes with key findings and policy recommendations for enhancing human rights protections in AI governance while fostering regional cooperation and technological innovation.

2. Literature Review

2.1. Human Rights Principles Relevant to AI Systems

The deployment of AI technologies raises major human rights concerns (Agalliu & Hoxha, 2024). Privacy and data protection are primary issues, as AI requires extensive personal data. Human Rights Watch (2023) highlights risks to privacy from surveillance and automated decision-making, stressing the need for safeguards. The right to privacy is protected by instruments such as the European Convention on Human Rights and the EU Charter, which are relevant for Western Balkan countries in the EU accession stages. Nondiscrimination and equality are also critical, with evidence of algorithmic bias in AI applications. As Bajrami and Halili (2024) note, while Western Balkan economies have ratified key human rights conventions, few have provisions against algorithmic discrimination, raising concerns given the region's ethnic and political divisions. Freedom of expression and information is another major consideration in AI governance. Content moderation algorithms and filtering mechanisms impact information flow and expression. Additionally, the SHARE Foundation (2024) argues that digital rights in the Western Balkans face challenges from regulatory gaps and automated content systems, raising concerns about censorship and free expression. Transparency and accountability are essential human rights dimensions for AI systems. The *"black box"* nature of AI complicates decision-making transparency and accountability for harmful outcomes. For example, the Institute for Technology and Society (2024) highlights Kosovo's lack of AI governance bodies, leading to accountability deficits, transparency and explainability with respect to the boundaries between AI usage and human rights.

2.2. Global AI Regulation Frameworks and Their Human Rights Dimensions

The EU's AI regulatory framework significantly impacts the Western Balkans' EU integration aspirations (see the EU Act, 2023). The EU AI Act establishes a risk-based framework to mitigate the potential harm of AI, aiming to protect human rights while fostering innovation (see the EU Act, 2023). Complementary regulations, such as the DSA and DMA, address broader digital governance concerns (see the Open Society Foundation for the Western Balkans,2024). However, a recent analysis highlights that while EU citizens are protected by these regulations, Western Balkan citizens lack equivalent protections, creating challenges for consistent human rights safeguards across Europe (see European Western Balkans, 2025). The Council of Europe has established frameworks, including Convention 108+ and its protocols for AI guidelines, focusing on human oversight, transparency, and nondiscrimination (see Convention 108+ and its protocols, 2018). In her thesis, Sollaku (2023) highlights that these principles are not fully adopted in the Western Balkans and that the region faces many challenges in AI advancement. The Council's human rights offer standards for regional assessments, complemented by UN frameworks that outline corporate responsibilities and privacy concerns in technology deployment (see the UN Recommendation on the Ethics of Artificial Intelligence, 2021). These global benchmarks are essential for evaluating regional AI governance and human rights protection.

2.3. Hermeneutical Theory in Legal Analysis

Hermeneutics, the theory of interpretation, offers insights into how legal texts and human rights principles are contextualized (Stelmach, 2023). Developed initially in the literature, it now informs legal analysis, particularly in AI regulation, by examining meaning construction in specific sociocultural contexts. Key hermeneutical concepts, such as the *"hermeneutic circle"*, highlight the interplay between parts and the whole and the influence of preunderstanding on interpretation (Stelmach, 2023). As AI challenges traditional legal frameworks, hermeneutics aid in adapting these frameworks to new contexts. Regional courts, such as the European Court of Human Rights, are crucial in shaping hermeneutical interpretations of human rights relevant to AI governance in the Western Balkans (Erdogan, 2021).

2.4. Regional Legal Frameworks in the Western Balkans

The Western Balkans are harmonizing their legal systems with EU standards, progressing variably. The EU accession process promotes AI governance reforms, particularly in terms of data protection. Krivokapić et al. (2023) emphasized the need to strengthen institutional capacities, presenting both opportunities and challenges for AI governance frameworks. Regional bodies such as the Regional Cooperation Council aid coordination, but progress remains mostly national. Legal traditions affect rights protection and regulation; Serbia is advancing AI regulation with a new 2024-2030 strategy (Schoenherr, 2024), whereas Kosovo lacks a comprehensive framework (ITS, 2024). Variations result from differing institutional capacities and political priorities. All countries in the region have adopted data protection laws aligned with the EU's GDPR, but implementation is inconsistent, and privacy safeguards related to AI are underdeveloped (see European Parliament, 2020). Gaps remain in understanding human rights principles in AI governance. Existing research often neglects hermeneutical aspects of legal texts and human rights. This study aims to address this gap by analyzing human rights protections in AI regulation through a hermeneutical lens across the Western Balkans.

3. Methodology

This study uses a hermeneutical approach to analyze AI systems' human rights protections in the Western Balkans. It outlines the methodology, data sources, analytical framework, and study limitations. Figure A provides a comprehensive view of the methodological framework and analysis employed in this research.

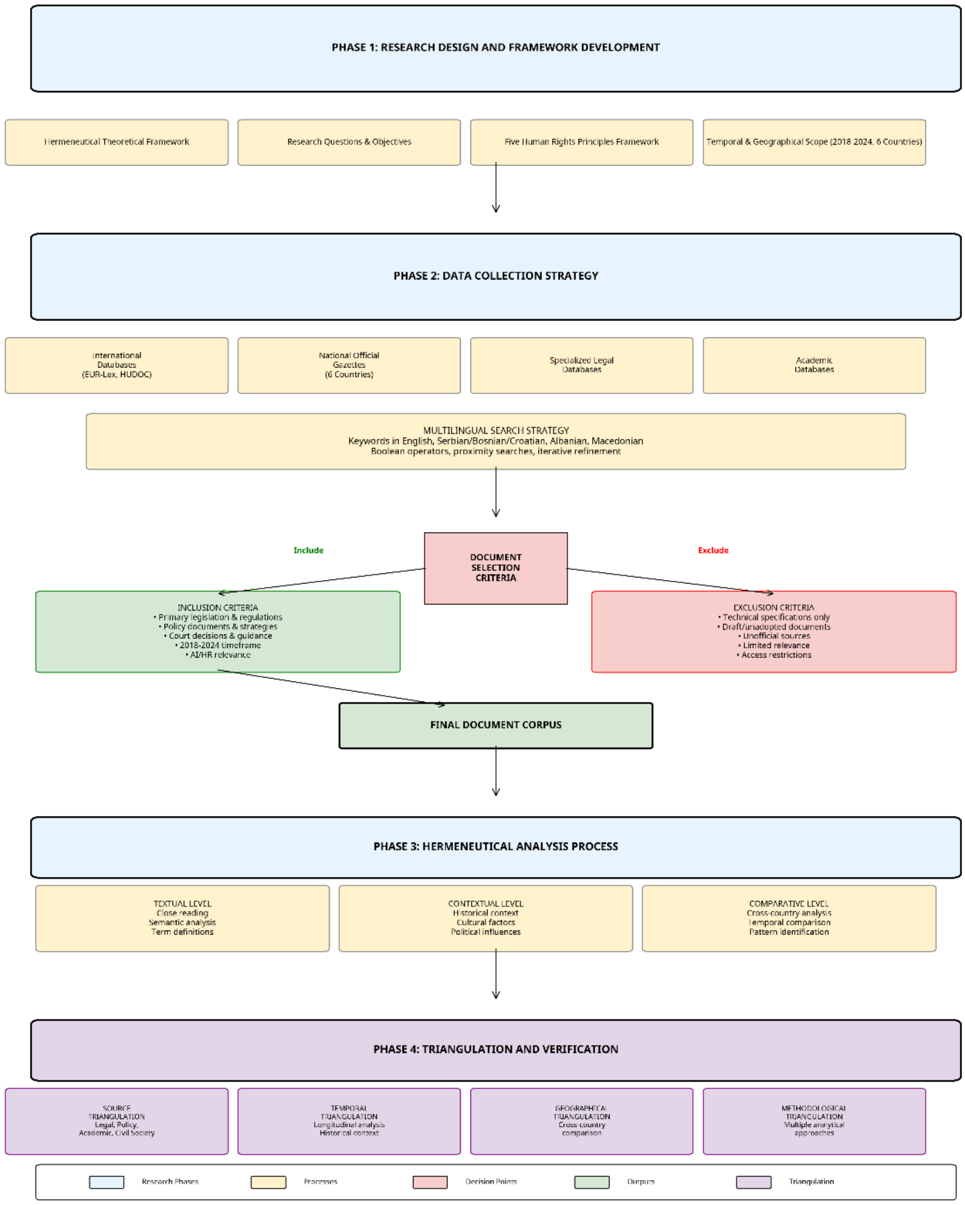

Figure 1 Methodological framework for hermeneutical analysis of AI governance and human rights in the Western Balkans.

3.1. Research Design and Hermeneutical Framework

Our research adopts a hermeneutical methodology grounded in Gadamer's philosophical hermeneutics, which emphasizes the interpretative nature of understanding legal and policy texts within their historical and cultural contexts. The hermeneutical approach is particularly suited to analyzing how human rights principles are understood, interpreted, and implemented in the context of AI governance across the culturally and politically diverse Western Balkans region. The research design incorporates three levels of hermeneutical analysis. The textual level involves close reading and semantic analysis of legal and policy documents, focusing on how key concepts such as "privacy," "transparency," "accountability," and "nondiscrimination" are defined and operationalized in AI governance contexts. The contextual level situates these texts within their broader historical, cultural, and political frameworks, acknowledging how factors such as socialist legacies, postconflict transitions, and EU integration aspirations influence the interpretation of human rights principles. The comparative level explores differences and similarities among the six Western Balkan countries, identifying regional patterns and variations in conformity with international human rights standards.

3.2. Hermeneutical Analysis Approach

The analysis emphasized three interpretative levels. First, textual "transparency" and "accountability" in AI governance documents. Second, contextual analysis situates these texts within their historical, cultural, and political frameworks, acknowledging the influence of factors such as socialist legacies and EU integration on human rights interpretations. Third, comparative analysis explores differences among the six Western Balkans countries, Serbia, Bosnia and Herzegovina, North Macedonia, Albania, Montenegro, and Kosovo, identifying regional patterns and varying conformity with international human rights standards, with a focus on both legal frameworks and practical implementations. The study refers to Gadamer's philosophical hermeneutics, emphasizing preunderstanding and historical consciousness in interpretation (Regan, 2012). It explores how Western Balkans countries interpret and implement human rights principles in AI governance, highlighting regional understanding and practical implications.

3.3. Temporal and geographical scope

3.3.1. Temporal Framework

The primary analysis period spans from January 2018 to December 2024, a six-year timeframe strategically selected to capture the evolution of AI governance frameworks in the Western Balkans. The starting point of 2018 was chosen because it marks several significant developments: the implementation of the EU General Data Protection Regulation (GDPR) in May 2018, which significantly influenced data protection legislation across the Western Balkans; the beginning of serious AI policy discussions in the region; and the initiation of Serbia's first AI strategy development process. The ending point of December 2024 ensures the inclusion of the most recent policy developments and legislative changes, captures the impact of EU AI Act adoption on regional policy discussions, and provides sufficient temporal distance for meaningful hermeneutical analysis. For contextual understanding, we also reviewed key foundational documents from 2010-2017, including constitutional provisions related to privacy and human rights, early data protection legislation, EU accession negotiation documents, and regional cooperation agreements.

3.3.2. Geographical Coverage

The study encompasses all six Western Balkan countries: Serbia, Bosnia and Herzegovina, North Macedonia, Albania, Montenegro, and Kosovo. This comprehensive regional approach allows for systematic comparison while acknowledging the unique political, legal, and cultural contexts of each country. The selection includes both EU candidate countries (Serbia, North Macedonia, Albania, Montenegro) and potential candidates (Bosnia and Herzegovina, Kosovo), providing insights into how different stages of EU integration affect AI governance approaches.

4. Data Sources

This research employs diverse data sources for hermeneutical analysis, focusing on secondary sources such as academic journals and legal documents regarding AI governance and human rights in the Western Balkans. Key materials include national legislation on AI governance, national AI strategies, regulatory guidelines, constitutional provisions, and regional agreements. The academic literature provides theoretical and comparative perspectives, whereas regional reports offer insights into the practical implementation of regulatory frameworks through civil society assessments, international organization reports, EU progress reports, technology sector analyses, and case studies on AI deployment and its implications for human rights.

4.1. Legal databases and official sources

The study collection strategy employed a systematic approach to accessing legal and policy documents across multiple database types and official sources. The comprehensive database access strategy included the following categories:

- International and Regional Legal Databases: EUR-Lex (European Union Law Database) was accessed for EU legislation and policy documents relevant to AI regulation, including the EU AI Act, Digital Services Act, and related directives.

- The Council of Europe Treaty Office provided access to Convention 108+ and related protocols on data protection and privacy.

- The European Court of Human Rights HUDOC database was searched for relevant jurisprudences on privacy, technology, and human rights.

- The United Nations Treaty Collection was consulted for international human rights instruments ratified by the Western Balkan countries.

- National Official Gazettes and Legal Repositories: The official legal publication systems of several countries were accessed systematically. Serbia used the official gazette of the Republic of Serbia, whereas Bosnia and Herzegovina required both state- and entity-level publications. North Macedonia, Albania, Montenegro, and Kosovo each had their own official gazettes, which provided access to national legislation and regulations.

- Specialized Legal and Administrative Databases: This study examined legislative processes and debates through national parliamentary websites and databases, consulted data protection authority materials, and reviewed court decisions on technology and privacy.

- Academic and Research Databases: Westlaw International, HeinOnline, JSTOR, and Google Scholar were used to access legal analysis, historical documents, interdisciplinary research, and recent academic publications.

Table 1 provides a summary of the referred sources and their accession periods:

Table 1 Summary of the database access.

Database Type	Specific Databases	Countries Covered	Document Types	Access Period
International	EUR-Lex, HUDOC, UN Treaty Collection	All 6 All 6 countries	Legislation, jurisprudence, treaties	2018-2024
National Official	Official Gazettes (6 countries)	Individual countries	Laws, regulations, decrees	2018-2024
Specialized Legal	Parliamentary websites, court databases	Variable by country	Debates, decisions, guidance	2018-2024
Academic	Westlaw, HeinOnline, JSTOR	Regional coverage	Academic literature, analysis	2010-2024

4.2. Search Strategy and Keywords

The search strategy was developed iteratively to cover the diverse languages of the Western Balkans and included multiple phases for refinement and expansion on the basis of initial results and expert input. The primary search terms in English combine artificial intelligence (AI) with human rights concepts, covering areas such as AI governance, regulation, algorithmic decision-making, facial recognition, data protection, privacy, nondiscrimination, transparency, and accountability. Searches were conducted in local languages, including Serbian, Bosnian, Croatian, Montenegrin, Albanian, and Macedonian, via equivalent key terms. Advanced techniques, such as Boolean, proximity searches, wildcards, date limiters, and document type filters, were employed to focus on legislation and policy documents. The search strategy was refined through multiple iterations. Broad initial searches identified regional terminology and legal language variations, which were refined with expert input for cultural and linguistic accuracy, culminating in comprehensive searches. A summary of these search criteria is presented in Table 2.

4.3. Document Selection Criteria

The document selection process used clear inclusion criteria to ensure the relevance and quality of the documents while covering the entire research area. It included various document types, such as primary legislation, national strategies, policy documents, court decisions, and parliamentary debates. These sources helped analyze AI governance, privacy, human rights, and technology regulations. The geographic focus was on the six Western Balkan countries, incorporating relevant EU and Council of Europe documents. Priority was given to materials directly addressing artificial intelligence, algorithms, and data protection issues. Documents were chosen from languages such as English, Serbian, and Bosnian, with official translations prioritized. This approach offered a comprehensive view of the legal and policy landscape related to AI in the specified region, emphasizing human rights and international standards.

Table 2 Search Terms by Language.

Language	Core AI Terms	Human Rights Terms	Governance Terms
English	artificial intelligence,	human rights, privacy, non discrimination	governance, regulation, framework
Serbian/ Bosnian/ Croatian	veštačka inteligencija, algoritam	ljudska prava, privatnost	upravljanje, regulacija
Albanian	inteligjenca artificiale, algoritëm	të drejtat e njeriut, privatësia	qeverisja, rregullimi
North Macedonian	вештачка интелигенција, алгоритам	човекови права, приватност	управување, регулација

4.4. Exclusion criteria

The exclusion criteria aim to ensure document quality and focus by removing irrelevant or unreliable sources. Excluded documents include purely technical specifications without legal or policy aspects, internal communications not publicly available, drafts never adopted, and academic commentaries treated separately. News articles and social media posts are also excluded because of concerns about reliability and authority. To ensure quality, documents from unofficial sources, inaccurate translations, and corrupted texts are excluded. Relevance is maintained by excluding documents that only briefly mention AI or lack specific technological focus.

The research process uses multiple triangulation strategies to ensure reliability and validity in qualitative research, particularly focused on hermeneutical analysis, which involves interpreting textual materials. Several forms of triangulation were used: *source triangulation* involved comparing information from various sources, such as legal, policy, administrative, international, academic, and civil society sources, to ensure comprehensive and balanced findings. *Temporal triangulation* tracked changes in AI governance related to human rights from 2018-2024, considering historical contexts and developmental patterns. *Geographical triangulation* compared data across six Western Balkan countries, identifying regional trends and external influences such as EU integration. *Methodological triangulation* uses different analytical methods, including textual, contextual, comparative, and interpretative analyses, to provide a deep understanding of how human rights principles are understood and applied in various contexts.

4.5. Validation procedures

The key findings were verified via diverse sources to ensure reliability, with contradictory information explored further. Gaps were analyzed through interpretation. Preliminary findings were discussed with experts in human rights and technology law, AI governance, and data protection. The analysis was refined by testing interpretations and verifying themes through multiple rounds of verification.

4.6. Analytical Framework

4.6.1. Human Rights Principles Framework

Our analytical framework centers on five key human rights principles governing artificial intelligence, providing a systematic structure for analyzing regional AI regulation strategies and their hermeneutical interpretation.

- Privacy and Data Protection: This principle examines how privacy rights are defined and protected in AI contexts, what mechanisms safeguard personal data in AI systems, and how regional interpretations compare to international standards. The analysis considers both formal legal protection and practical implementation challenges.

- Nondiscrimination and Equality: This dimension analyzes how algorithmic bias and discrimination are addressed in legal and policy frameworks, what protections exist for marginalized groups, and how regional approaches to equality translate into AI governance contexts.

- Transparency and Accountability: This principle investigates what criteria govern AI transparency requirements, how accountability mechanisms are structured, and what remedies are available for rights violations in automated decision-making contexts.

- Freedom of Expression and Information: This aspect examines how AI regulations balance innovation with free expression and information access, particularly in content moderation and information filtering contexts.

- Human Dignity and Autonomy: This principle analyzes how human agency and dignity are preserved in automated decision-making processes and what safeguards exist against dehumanizing technological applications.

This framework enables a systematic analysis of the recognition, interpretation, and operationalization of human rights principles in regulatory contexts.

4.6.2. Three-dimensional analysis

The study employs a three-dimensional analysis of human rights principles, which includes formal acknowledgment, an interpretative approach, and implementation mechanisms. Formal acknowledgment looks at legal recognition and status, the interpretative approach examines cultural and historical influences on AI governance, and implementation mechanisms explore strategies and enforcement effectiveness in practice.

4.7. Methodological Limitations

This research faced several limitations, including language barriers, limited AI governance jurisprudence, uneven source availability across countries, and resource constraints. To mitigate these issues, professional translation services were used for important documents, multiple sources were consulted, source availability was systematically documented, and iterative analysis and peer review were conducted to address resource limitations.

4.8. Ethical considerations

All research was conducted in accordance with established ethical standards for legal and policy research. Only publicly available documents were analyzed, ensuring compliance with access and privacy requirements. Proper attribution was provided for all sources and expert consultations. The research design avoided any potential harm to individuals or institutions while maintaining analytical rigor and critical assessment of policy frameworks.

5. Results and Discussion

5.1. Comparative Analysis of AI-Related Legislation in Western Balkans

The Western Balkans' AI regulatory framework is fragmented and varies in effectiveness. Serbia led with its 2019 Strategy for AI Development (see the Strategy for the Development of Artificial Intelligence in the Republic of Serbia, 2019), recognizing human rights through existing laws. North Macedonia initiated AI governance under its National ICT Strategy (see North Macedonia National ICT Strategy, 2021) but lacked a dedicated framework and took a reactive approach to human rights. Montenegro, an EU candidate, aligns digital governance with EU standards in its Digital Transformation Strategy but lacks specific human rights protections. Albania's AI governance is nascent, with limited legal recognition and minimal guidance on human rights protections in its Digital Agenda. As Sollaku (2023) highlights in her thesis, *"significant gaps in addressing fundamental rights implications of algorithmic systems"*. Bosnia and Herzegovina face challenges due to their fragmented governance and lack of state-level digital coordination, which leads to inconsistent entity-level approaches (Krivokapić et al., 2023). Kosovo, the youngest state, lacks dedicated AI laws or governance, creating a regulatory vacuum, although it has modern data protection aligned with the GDPR (see the ITS study, 2024).

5.1.1. Data Protection Laws

Data protection frameworks are the most developed AI governance regulations in the Western Balkans, with all six countries adopting laws aligned with the EU's GDPR, although implementation varies (Ruzic, 2021). They safeguard privacy rights in AI, such as purpose limitations and data minimization. For example, Serbia's 2018 law on the GDPR closely mirrors and protects against significant automated decision-making effects, but its effective implementation is limited (Spalević & Vićentijević, 2022). Kosovo's law also includes GDPR principles, but challenges undermine effectiveness (Zejnullahu, 2022). The literature also reveals that although legally established, these frameworks are insufficient for broader human rights issues related to AI, lacking provisions for machine learning challenges such as inferred data and algorithmic transparency (Ruzic, 2021).

5.1.2. Digital Governance Frameworks

Digital governance frameworks in the Western Balkans offer various human rights protections in technology, including digital strategies, electronic communication regulations, and cybersecurity provisions relevant to AI. For example, Serbia's laws on electronic communications and information security provide mechanisms for technology governance focused on security and reliability (see Law on Electronic Communications, 2014). However, as noted by the SHARE Foundation (2024), these lack provisions for the unique challenges of AI, such as transparency and accountability (see the SHARE Foundation, 2024). The European Western Balkans (2025) emphasize a regulatory gap: *while EU citizens benefit from the Digital Services Act (DSA) and similar regulations, citizens in the Western Balkans do not, complicating consistent human rights protections regionally*. Furthermore, the report of the Open Society Foundations for the Western Balkans (2024) states that the EU AI Act aims to manage risks to fundamental rights, a risk-based approach not yet integrated into Western Balkans regulations, potentially leaving human rights protections vulnerable as AI is deployed.

5.1.3. Sectoral Regulations

Sectoral regulations in areas such as healthcare, finance, and public administration provide additional layers of governance for AI deployment across the Western Balkans. These domain-specific frameworks often include provisions on confidentiality, consent, and service standards that have implications for how AI systems can be deployed in these contexts. In the healthcare sector, regulations on medical data confidentiality and patient rights establish important parameters for AI applications in diagnostics, treatment planning, and health administration. For example, Serbia's Law on Health Care (see Law on Health Care, 2019) includes provisions on patient privacy and medical confidentiality that apply to technological applications, although without specific reference to AI systems. Similar frameworks exist across the region, although with limited explicit recognition of AI-specific challenges in healthcare contexts. Financial sector regulations across the Western Balkans include provisions on algorithmic trading, credit scoring, and fraud detection that are relevant for AI applications in these domains. Montenegro's financial services regulations, for example, include requirements for transparency and explainability in credit decisions that provide some protection against discriminatory or opaque algorithmic assessments (Ruzic, 2021). However, these protections remain limited and inconsistent across the region. Public administration represents a particularly important sector for AI governance, as governments across the Western Balkans increasingly deploy algorithmic systems for service delivery, resource allocation, and decision-making (Borici et al., 2025). Serbia's Law on Electronic Government (cited in Golic, 2023) establishes principles for the digitalization of public services, including requirements for accessibility and nondiscrimination. However, these frameworks lack specific safeguards for algorithmic decision-making in public administration, creating risks for transparency, accountability, and equal treatment (Krivokapić et al., 2023).

5.2. Gaps and challenges in human rights protection

The comparative analysis of legal and policy frameworks reveals several significant gaps and challenges in human rights protections related to AI across the Western Balkans. First, there is a notable absence of AI-specific legislation that addresses the unique challenges posed by these technologies. While general legal frameworks provide some protection, they often fail to address novel issues such as algorithmic transparency, explainability, and accountability. Second, there are significant implementation gaps between formal legal protections and practical realities. Even where relevant legal frameworks exist, limited institutional capacity, resource constraints, and enforcement challenges undermine their effectiveness in practice. The ITS (2024) study on Kosovo highlights that "the gap between legal frameworks and implementation capacity creates significant risks for human rights as AI systems are deployed without adequate oversight or safeguards." Third, there is limited regulatory attention to algorithmic discrimination and bias, despite the region's complex ethnic, religious, and political divisions. While all Western Balkan countries have adopted antidiscrimination legislation, these frameworks rarely address the specific challenges of algorithmic bias or provide mechanisms for identifying and addressing discriminatory outcomes in AI systems. Fourth, there are significant disparities in regulatory development across the region, creating risks for inconsistent human rights protections. While countries such as Serbia have made progress in developing AI governance frameworks, others such as Kosovo, Bosnia and Herzegovina lag significantly behind, creating a fragmented regional landscape for human rights protections in AI contexts. Fifth, there is limited regional coordination in AI governance, despite the cross-border nature of many AI applications and services. The absence of harmonized approaches creates challenges for consistent human rights protections across the region and complicates alignment with EU standards as part of the accession process. Table 1 provides a comparative overview of key legal provisions related to AI governance across the six Western Balkans countries, highlighting significant variations in regulatory development and human rights protections.

5.3. Cultural and Societal Interpretations of Human Rights

The interpretation and implementation of human rights principles in AI governance across the Western Balkans are shaped not only by formal legal frameworks but also by distinctive cultural and societal contexts. The Western Balkans region has distinctive cultural values and historical experiences that shape interpretations of key human rights principles relevant to AI governance. Privacy, for example, is understood through lenses influenced by communal traditions, historical experiences of surveillance, and more recent digital transformations (Rusic, 2021). These dynamics are evident across the region, creating distinctive interpretative contexts for privacy rights in AI governance. Serbia's relative emphasis on state security in its regulatory approach reflects historical experiences and political priorities that shape how privacy is balanced against other considerations (Spalević & Vićentijević, 2022). Montenegro's coastal tourism economy has influenced its approach to data governance, with greater attention to international standards and cross-border data flows reflecting economic priorities (see the EU report for Montenegro, 2025). Conceptions of human dignity across the Western Balkans are similarly influenced by distinctive cultural and historical factors. Religious traditions, including Orthodox Christianity, Catholicism, and Islam, have shaped the understanding of human dignity in ways that influence approaches to technological governance (Sollaku, 2023). These traditions often emphasize human uniqueness and the spiritual dimension of personhood, creating potential tensions with increasingly automated decision-making systems. Equality and nondiscrimination principles are interpreted through the complex lens of the region's ethnic, religious, and political divisions (see FRAME, 2017). The legacy of conflict in the 1990s created heightened sensitivity to discrimination along ethnic and religious lines, although this has not consistently translated into robust protection against algorithmic discrimination. As Krivokapić et al. (2023) observe, *"while anti-discrimination frameworks exist across the Western Balkans, their application to algorithmic contexts remains underdeveloped, creating risks that AI systems could reinforce existing societal divisions."* Table 3 gives a comprehensive view of the AI regulations and Human Rights safeguards in the Western Balkans.

Table 3 Comparative Analysis of AI Regulation in Western Balkans.

Country	Dedicated AI Strategy	Data Protection Law with Automated Decision-Making Provisions	Algorithmic Impact Assessment Requirements	Explicit Human Rights Safeguards in AI Contexts
Serbia	Yes (2020-2025)	Yes	Limited	Partial
Bosnia and Herzegovina	No	Yes	No	No
North Macedonia	No (only in ICT Strategy)	Yes	No	No
Albania	No	Yes	No	No
Montenegro	No (only in Digital Transformation Strategy)	Yes	No	Partial
Kosovo	No	Yes	No	No

Note: This table represents the regulatory landscape as of May 2025 on the basis of the sources cited in this research paper.

5.4. Post-Socialist Transition and Its Impact on Rights Discourse

The shift from socialist systems to market economies and democratic governance has significantly influenced rights discourse in the Western Balkans, creating unique contexts for human rights in technological governance shaped by socialist legacies (Poulou, 2014). Socialist legal systems in Yugoslavia and Albania prioritized economic and social rights while limiting political liberties, a legacy that affects current rights conceptualization with implications for AI governance (Poulu, 2014). The SHARE Foundation (2024) noted that these transitions led to hybrid rights frameworks blending socialist protections with individual liberties and market freedoms. Post-socialist privatization, especially in telecommunications and media, introduced regulatory challenges and altered the balance between state and private entities in digital governance, impacting responsibility for human rights protections in AI (Dalakoglou, 2012). Democratic consolidation has underscored the importance of the rule of law, transparency, and accountability in AI governance. However, as Krivokapić et al. (2023) highlight, Western Balkan economies often lack institutional capacity, raising vulnerabilities in human rights protections as AI systems are implemented.

5.5. Public Perception of Technology and Surveillance

Public perceptions of technology and surveillance in Western Balkans reflect historical experiences and current realities (Budak et al., 2014; Budak et al., 2012). The legacy of state surveillance during socialism shaped attitudes toward monitoring and data collection, affecting perceptions of AI surveillance systems (Budak et al., 2012; Hallinan et al., 2012). The latest ITS (2024) study in Kosovo noted that public attitudes toward surveillance technologies reflect tensions between security concerns, privacy values, and trust in institutions. This regional dynamic shows divided opinions on balancing AI security applications with privacy protections. Trust in institutions is crucial for shaping public perceptions of AI governance. In the Western Balkans, low levels of trust in government compared with EU averages hamper the establishment of legitimate AI governance frameworks. Furthermore, the SHARE Foundation (2024) noted that *"limited public trust in regulatory institutions undermines the perceived legitimacy of technological governance frameworks."* Media coverage and public discourse on AI across the region tend to emphasize either economic opportunities or security applications, with limited attention to human rights implications (SHARE Foundation, 2024). This framing shapes public understanding and expectations regarding appropriate governance frameworks. The relative absence of robust civil society engagement in AI governance issues in some countries further limits public discourse on the human rights dimensions of these technologies (Skolnik & Haman, 2024).

6. Hermeneutical interpretation of legal texts and policies

6.1. Textual Analysis of the Key Legal Provisions

A close reading of legal texts in the Western Balkans reveals significant variations in defining key AI governance concepts (Boriçi et al., 2025). Terms such as *"automated decision-making," "algorithm," "artificial intelligence," and "profiling"* are inconsistently *defined as "profiling"* as any automated processing of personal data to evaluate the personal aspects of individuals, closely mirroring GDPR language but lacking AI-specific limitations (see Law on Personal Data Protection of RS, 2019). Kosovo's law similarly addresses automated decision-making with different terminology and insufficient implementation guidance for AI (see Law on Protection of Personal Data of Kosovo, 2019). The lack of detailed guidance contributes to ambiguity in applying protection to AI in Kosovo. The legal texts in the region vary in technological specificity, predominantly using technology-neutral language that may inadequately address AI challenges, leading to potential interpretative gaps as technologies advance.

6.2. Interpretative Approaches in Judicial and Administrative Practice

Judicial and administrative interpretations of legal texts offer insights into human rights principles in AI governance (Hogan & Lasek-Markey, 2024). While AI-specific jurisprudence in Western Balkans is limited, general legal interpretations in technology provide valuable perspectives. Data protection authorities in the region address AI issues through guidance and enforcement, although their effectiveness varies (Boriçi et al., 2025). Serbia's Commissioner for Information has issued key opinions on biometric surveillance that balance security and privacy rights (see Law on Personal Data Protection of RS, 2019). Kosovo's Information and Privacy Agency faces capacity constraints that hinder detailed guidance on AI applications, as noted in the ITS (2024) study, which highlights that limited resources affect interpretations of data protection in complex technological contexts. Administrative practices in public procurement and AI deployment highlight interpretative patterns. Public authorities often interpret legal requirements narrowly, neglecting robust human rights impact assessments and transparency in algorithmic systems, which may weaken legal protection. The lack of judicial decisions on AI creates interpretative gaps; as Krivokapić et al. (2023) state, *"the lack of jurisprudence on AI-specific human rights challenges leaves significant ambiguity in how existing legal frameworks apply to novel technological contexts."* Thus, interpretation relies heavily on administrative practice and policy implementation, which often lack transparency and accountability.

6.3. Divergence from International Human Rights Standards

The hermeneutical analysis indicates significant divergences between regional human rights interpretations in AI and international standards, shaped by explicit policy choices and implicit patterns in the Western Balkans. The principle of transparency in algorithmic systems exemplifies these divergences. International standards highlight meaningful transparency in automated decision-making, whereas Western Balkan legal frameworks generally lack specific transparency requirements, creating gaps between international norms and regional practices (Hoxhaj, 2020). The principle of nondiscrimination in AI systems also involves various regional interpretations. While international standards call for proactive measures against algorithmic bias, regional frameworks typically adopt reactive approaches that inadequately address discrimination challenges (cited in Krivokapic et al., 2023). Finally, the right to an effective remedy for rights violations in AI contexts is another area of divergence. International standards stress accessible remedies for automated system violations; however, as noted by the SHARE Foundation (2024), remedial mechanisms in the Western Balkans remain underdeveloped, resulting in accountability gaps in AI governance. Divergences from international standards stem from capacity constraints and regional interpretations of human rights. EU integration processes may pressure alignment with European standards, potentially reducing these divergences. However, the European Western Balkans (2025) analysis reveals persistent gaps between EU protections and regional implementations, posing challenges for human rights protections in AI contexts. Figure 2 illustrates a model of the interpretative layers shaping AI governance in the Western Balkans, highlighting the interaction of international standards, regional contexts, and national implementations in the formation of unique frameworks.

6.4. Case studies of AI deployment

6.4.1. Facial recognition systems in public spaces

Facial recognition technologies in the Western Balkans pose significant challenges for human rights in AI governance, raising issues of privacy, surveillance, consent, and proportionality (see OECD, 2021). Serbia's "Safe City" initiative in Belgrade employs extensive facial recognition systems supplied by Huawei, including numerous surveillance cameras cited in (Korac, 2021). In this project, however, civil society organizations contested the legal basis for this deployment, citing inadequate privacy safeguards. According to the SHARE Foundation (2024), Serbia's implementation lacked comprehensive impact assessments and public consultation, emphasizing security benefits over privacy, thus shaping human rights interpretations in favor of security objectives. Kosovo has explored facial recognition technologies for border control and law enforcement purposes, although with more limited implementation than Serbia has see (KOHA, 2024). The ITS (2024) study additionally revealed that *"Kosovo's consideration of facial recognition technologies has occurred in a significant regulatory vacuum, with limited explicit legal frameworks governing biometric surveillance."* This gap creates risks for arbitrary or disproportionate implementations that may undermine human rights protections. Montenegro has implemented facial recognition systems at border crossings, justified through security and tourism management objectives. The legal framework relies on border security legislation rather than AI-specific governance, which poses risks to human rights safeguards. Key patterns include prioritization of security objectives over privacy, inadequate governance of biometric surveillance due to a lack of AI-specific laws, and limited transparency and public consultation hindering democratic oversight.

Figure 2 Interpretative Layers in Western Balkans AI Governance.

Note: This figure shows the three main layers of AI governance in the Western Balkans, illustrating the filtering of international standards through regional contexts and their varied national implementation.

6.4.2. Algorithm decision-making in public services

The algorithmic systems in public service delivery in the Western Balkans highlight crucial issues of human rights protection in AI governance, particularly concerning transparency, nondiscrimination, due processes, and accountability (Hoxhaj, 2020). Social welfare systems in the region have adopted algorithms for eligibility, benefit calculations, and fraud detection (Krivokapic et al., 2023). Kosovo's automated social assistance program raises concerns about transparency and accountability in these assessments (see ITS, 2024). The ITS (2024) study noted that *"Kosovo's algorithmic social welfare systems lack transparency regarding decision criteria and appeal mechanisms, risking due process and accountability,"* prioritizing efficiency over transparency. Serbia's algorithmic employment services also face issues of discrimination and transparency in job matching. Krivokapić et al. (2023) state that *"the lack of nondiscrimination safeguards in employment algorithms risks reinforcing societal inequalities through automated decisions."* North Macedonia has piloted algorithmic systems for educational administration, raising concerns about equality, transparency and human rights protections due to a lack of legal safeguards. Key patterns include prioritization of efficiency over transparency, absence of algorithmic impact assessments, and limited public awareness hindering informed consent.

6.4.3. Cross-border data flows and regional cooperation

The cross-border movement of data for AI development is a key case study for human rights protections in regional AI governance, raising issues of jurisdiction and accountability (Shtupi & Prifti, 2025). Regional initiatives such as the Open Balkan and the Berlin Process promote economic integration and data sharing, but legal standards for AI data remain unclear across borders. The European Western Balkans (2025) report highlights that while the EU has established frameworks such as the GDPR, the Western Balkans lack harmonized approaches, leading to regulatory gaps for AI. This disparity risks inconsistent protection for the data used in AI systems. Commercial AI services, particularly from multinational companies, complicate the protection of human rights, with unclear legal standards when providers operate outside the region (Shtupi & Prifti, 2025). The Open Society Foundations Western Balkans (2024) report noted that the lack of harmonization in AI governance poses significant challenges for consistent rights protections in cross-border contexts. Several interpretative patterns emerge in cross-border contexts. Economic integration often prioritizes harmonized rights protections, creating regulatory gaps. Weak regional coordination hinders consistent human rights approaches in AI governance. External actors such as the EU and multinational tech companies shape the interpretation and implementation of rights principles. Figure 3 maps significant AI deployments in the Western Balkans, showing their geographical distribution and sectoral focus.

6.4.4. Challenges and opportunities

The hermeneutical analysis of human rights protections in AI governance across the Western Balkans reveals significant challenges and opportunities for strengthening rights-centered approaches to technological regulation. Western Balkans countries struggle to balance technological advancement with human rights protections (Borici et al., 2025). Economic priorities push for rapid AI adoption, potentially outpacing regulatory frameworks. Serbia's Strategy for the Development of Artificial Intelligence (see the AI Development Strategy of RS, 2019) seeks regional leadership in AI, prioritizing economic opportunities and innovation, but may conflict with comprehensive rights protections. Krivokapić et al. (2023) noted that the focus on AI for development risks sidelining human rights. The belief that strong regulation hampers innovation influences the interpretation of human rights in AI governance, leading to limited protection and slow regulatory responses. The challenge is creating governance that safeguards rights while fostering innovation, requiring nuanced frameworks and institutional capacity. The EU integration process provides important external pressure for balancing development with rights protections, as alignment with EU standards becomes increasingly important for accession countries. The European Western Balkans (2025) analysis noted that *"the EU's rights-centered approach to AI regulation, embodied in the AI Act, creates significant pressure for Western Balkans countries to develop more comprehensive human rights protections in their AI governance frameworks."* This external influence may help shift the balance toward stronger rights protections over time.

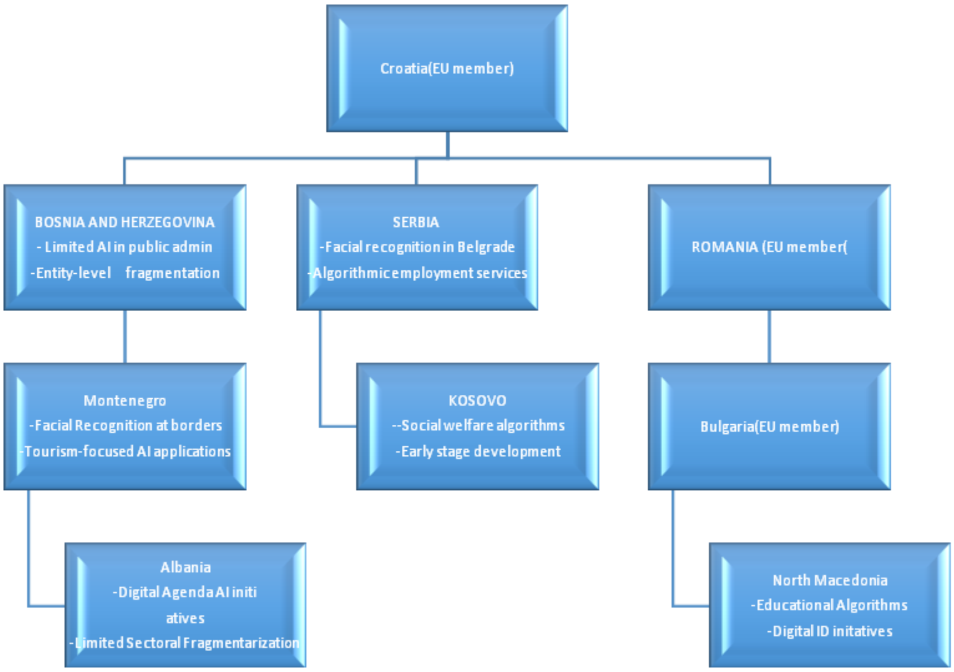

Figure 3 Map of AI Deployment Cases in the Western Balkans.

Note: This figure maps key AI deployment cases across the Western Balkans region, highlighting significant applications in each country and their sectoral focus. The map illustrates the uneven development of AI implementation across the region, with Serbia showing more advanced applications, while countries such as Kosovo, Bosnia and Herzegovina remain at earlier stages.

6.4.5. Capacity gaps in regulatory oversight

Significant capacity gaps in regulatory oversight pose a major challenge to human rights protections in AI governance across the Western Balkans (Borici et al., 2025). Limited resources, expertise, and enforcement capabilities hinder the effectiveness of legal frameworks, risking rights protections as AI systems are implemented. Data protection authorities struggle with capacity issues specific to AI. The ITS (2024) study by Kosovo noted that "limited staffing, technical expertise, and financial resources constrain the Information and Privacy Agency's ability to provide effective oversight of complex AI applications." Similar constraints affect regional regulatory bodies, resulting in gaps between formal protections and practical implementation. The lack of technical expertise is a critical gap, as regulatory authorities often lack knowledge of AI systems and their human rights implications. This knowledge deficit hampers oversight, guidance development, and the enforcement of protection. The SHARE Foundation (2024) states that "*the technical complexity of AI systems creates significant challenges for regulatory authorities with limited specialized expertise*." Enforcement capabilities also pose challenges, as authorities often lack the tools and resources to ensure compliance with human rights protections in AI (the SHARE foundation, 2024). Limited penalties, procedural barriers, and jurisdictional issues further impede enforcement effectiveness, threatening the translation of formal protections into practical safeguards. Addressing capacity gaps necessitates investment in institutional development and training.

The EU accession process offers resources for capacity building, but challenges in the regulatory oversight of complex technologies persist.

6.4.6. Regional Cooperation Potential

Given their shared experiences, geographical proximity, and EU integration aspirations, significant opportunities for enhancing human rights protection through regional cooperation on AI governance exist in the Western Balkans. Current mechanisms such as the Regional Cooperation Council and the Western Balkans Fund can foster unified approaches, as evidenced by initiatives supporting research on AI and human rights (see RCC, 2025). The Berlin process initiative potentially enhanced coordinated digital governance despite its limited coverage (see the Berlin process, 2025). As noted by Krivokapić et al. (2023), "*regional economic integration initiatives create both opportunities and challenges for AI governance*." EU integration provides external frameworks and encourages alignment with EU standards, leading to potential regulatory convergence, although national variations persist. The European Western Balkans (2025) report highlights that *"the EU's comprehensive approach to digital governance... provides important reference points for regional cooperation on human rights protections."* Enhanced regional cooperation could address key challenges such as cross-border data flows, regulatory gaps, and harmonized interpretations of human rights in AI (Tikos & Krasznay, 2022). However, achieving this requires political will, institutional frameworks, and resources potentially hindered by competing priorities and regional tensions.

6.4.7. Future Trajectories for Human Rights-Centered AI Governance

Multiple potential trajectories for human rights-centered AI governance exist in the Western Balkans, reflecting varying balances of technological advancement, human rights, regional cooperation, and international standards. The EU integration pathway is the primary external influence on AI governance, with countries aligning with EU frameworks such as the AI Act and GDPR as they advance in accession negotiations. This suggests increased convergence with European standards, although variations will persist across national contexts. A second trajectory is the development of national AI strategies, where countries, following Serbia's example, may incorporate explicit human rights protections.

The implementation of these strategies significantly affects the interpretation and safeguarding of human rights in AI contexts. Regional cooperation initiatives present a third avenue for harmonized AI governance, enhancing human rights protections and addressing fragmentation through standards and capacity building.

Civil society engagement and public discourse on AI ethics represent a fourth avenue that could pressure governance frameworks to prioritize rights protections and transparency over development.

These trajectories are interconnected, reflecting complex interactions influenced by political priorities, institutional capacities, external factors, and societal values in the Western Balkans.

7. Conclusions

This analysis of human rights protections in AI deployment in the Western Balkans highlights significant variations in regulatory approaches and capacities. Examining legal texts and cultural contexts reveals key findings for enhancing rights-centered technological regulation. A comparative analysis of six countries, Serbia, Bosnia and Herzegovina, North Macedonia, Albania, Montenegro, and Kosovo, revealed disparities in regulatory development. Serbia is developing AI governance frameworks, whereas Kosovo lacks dedicated laws, creating substantial regulatory vacuums that risk inconsistent human rights protections amid cross-border technological applications. Data protection frameworks are the most developed regulations for AI governance in the Western Balkans, with all six countries adopting laws aligned with the EU's General Data Protection Regulation. However, these frameworks often lack specific provisions for challenges posed by AI systems, particularly in terms of algorithmic transparency and accountability, creating vulnerabilities in human rights protections as technology evolves. A hermeneutical lens shows that cultural and societal contexts significantly influence the interpretation of human rights principles in AI governance. Regional values, postsocialist transitions, and public perceptions of technology shape the understanding of concepts such as privacy and equality, actively constituting the meaning of human rights principles in this context. Textual analysis of legal provisions reveals significant variations in the definitions and applications of terms such as "automated decision-making," "algorithm," and "profiling," creating challenges for consistent human rights protections, especially in cross-border contexts. Case studies on AI deployment, including facial recognition and algorithmic decision-making, show patterns prioritizing security over transparency and gaps between legal protections and implementation, posing risks to human rights. Additionally, capacity gaps in regulatory oversight, owing to limited resources and expertise, hinder effective legal frameworks and necessitate investment in institutional development and regulatory tools. Opportunities for strengthening human rights protections in AI governance exist in the Western Balkans. The EU integration process offers incentives for developing regulatory approaches aligned with European standards. Regional cooperation mechanisms can facilitate coordinated cross-border strategies. Civil society organizations are crucial in shaping discourse, monitoring implementation, and advocating for stronger protections. Effectively leveraging these opportunities could significantly enhance human rights as AI systems evolve. Policy recommendations include the following: Western Balkans countries should create AI governance frameworks with specific human rights safeguards, including algorithmic impact assessments, transparency in decision-making, and remedies for rights violations. Strengthening regional cooperation through coordination mechanisms, harmonized regulations, and capacity-building initiatives can address cross-border challenges and promote human rights protections, with the Regional Cooperation Council facilitating these efforts. Third, regulatory capacity must be strengthened through investment in resources, expertise, and enforcement tools. Data protection authorities need adequate staffing and specialized knowledge for the effective oversight of complex AI systems, with support from international partners such as the EU. Fourth, civil society engagement in AI governance should be expanded through funding for research, advocacy, and public education, enhancing oversight and accountability. Fifth, human rights impact assessments should be mandatory for significant AI deployments, particularly in the public sector, to address issues such as privacy and nondiscrimination with public consultation. The hermeneutical approach underscores the role of interpretative contexts in shaping human rights principles in AI governance, indicating that enhancing protection requires attention to both legal provisions and their contextual interpretations. As AI technologies evolve in the Western Balkans, the need for strong human rights protections becomes urgent. The region’s unique context, marked by postsocialist transitions, EU integration, and historical legacies, presents challenges and opportunities for rights-centered technological governance. This study aims to enhance AI governance approaches that protect fundamental rights while fostering technological development. Future research could explore how interpretations of human rights in AI governance evolve, particularly with respect to EU integration and new technologies. Studies may also examine the experiences of those impacted by AI systems, enhancing the understanding of rights protections in practice and the interplay between technology, regulation, and human rights.

Ethical considerations

All research was conducted in accordance with established ethical standards for legal and policy research. Only publicly available documents were analyzed, ensuring compliance with access and privacy requirements. Proper attribution was provided for all sources and expert consultations. The research design avoided any potential harm to individuals or institutions while maintaining analytical rigor and critical assessment of policy frameworks.

Conflict of Interest

The authors declare no conflicts of interest.

Funding

This research did not receive any financial support.

References

Agalliu, P., & Hoxha, A. (2024). *The European Union’s AI Act: Guarding against or Employing Artificial Intelligence*. https://migrationletters.com/index.php/ml/article/view/7615

Assembly of the Republic of Kosovo. (2019). *LAW NO. 06/L –082 ON PROTECTION OF PERSONAL DATA*. OFFICIAL GAZETTE OF THE REPUBLIC OF KOSOVO. https://www.dataguidance.com/sites/default/files/law_no._06_l082_on_protection_of_personal_data_0.pdf

Bajrami, E., & Halili, F. (2024). EXPLORING THE IMPACT OF ARTIFICIAL INTELLIGENCE ON THE IoT AND DIGITAL AGENDA IN THE WESTERN BALKANS: INTEGRATING a PROPOSED WEB APPLICATION FOR REGIONAL ADVANCEMENT. *Journal of Natural Sciences and Mathematics of UT*, *9*(17–18), 244-258. https://doi.org/10.62792/ut.jnsm.v9.i17-18.p2819

Berlin Process | Berlin Process. (2025). Berlin Process. https://www.berlinprocess.de/

Borici, N. A. (2025). Artificial intelligence in the Western Balkans: economic impact, ethical challenges, and regulatory frameworks. *Journal of Information Systems Engineering & Management*, *10*(39s), 224-230. https://doi.org/10.52783/jisem.v10i39s.7143

Budak, J., Anić, I., & Rajh, E. (2012). Public Attitudes towards Surveillance and Privacy in Western Balkans: The Case of Serbia. *The Institute of Economics Working Papers*, *3*, 5-22. http://hrcak.srce.hr/89368

Budak, J., Rajh, E., & Anić, I. (2014). Privacy concern in Western Balkan countries: Developing a typology of citizens. *Journal of Balkan and Near Eastern Studies*, *17*(1), 29-48. https://doi.org/10.1080/19448953.2014.990278

Commissioner for Information of Public Importance and Personal Data Protection. (2019). *L A W ON PERSONAL DATA PROTECTION* (No.97/08 107/12). Official Gazette of RS. https://www.poverenik.rs/images/stories/dokumentacija-nova/zakon-o-zastiti-podataka-o-licnosti_en.pdf

Council of Europe. (2018). *Convention 108 + Convention for the protection of Individuals with regard to the processing of personal data*. https://rm.coe.int/convention-108-convention-for-the-protection-of-individuals-with-regar/16808b36f1

Council, R. C. (2025). *Regional Cooperation Council | Welcome*. https://www.rcc.int/

Dalakoglou, D. (2012). ‘The Road from Capitalism to Capitalism’: Infrastructures of (Post)Socialism in Albania. *Mobilities*, *7*(4), 571-586. https://doi.org/10.1080/17450101.2012.718939

Erdoğan, M. (2021). The rise of hermeneutics in human rights interpretation in the Case-Law of the ECTHR and the domestic courts. *Annales De La Faculté De Droit D’Istanbul*, *0*(70), 91-118. https://doi.org/10.26650/annales.2021.70.0004

European Commission. (2017). *The role of human rights in the EU's external action in the Western Balkans and Turkey*. Fostering Human Rights among European Policies (FRAME). https://repository.gchumanrights.org/server/api/core/bitstreams/95055ada-f596-479f-9eb5-e86a292ddc14/content

European Parliament. (2020). *The impact of the General Data Protection Regulation (GDPR) on artificial intelligence*. European Parliamentary Research Service. https://www.europarl.europa.eu/RegData/etudes/STUD/2020/641530/EPRS_STU(2020)641530_EN.pdf

European Parliament. (2023, June). Artificial Intelligence Act (EU Legislation in Progress). European Union. https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/698792/EPRS_BRI(2021)698792_EN.pdf

European Union. (2025). *Artificial Intelligence Landscape Assessment (AILA) shaping AI to be an empowering force for people and planet Montenegro 2024*. https://www.undp.org/montenegro/publications/artificial-intelligence-landscape-assessment-aila

European Western Balkans. (2025, April 4). Why the Digital Services Act is needed in the Western Balkans: An institutional and market perspective. https://europeanwesternbalkans.com/2025/04/04/why-the-digital-services-act-is-needed-in-the-western-balkans-an-institutional-and-market-perspective/

Golić, D. (2023). Normative regulation of electronic administration in Republic of Serbia. *Pravo - Teorija I Praksa*, *40*(suppl), 4-62. https://doi.org/10.5937/ptp2300044g

Government of North Macedonia. (2021). *North Macedonia National ICT Strategy 2021–2025* (No. 2019/408197). https://ener.gov.mk/PublicDocuments/%D0%9D%D0%B0%D1%86%D1%80%D1%82%20%D0%9D%D0%B0%D1%86%D0%B8%D0%BE%D0%BD%D0%B0%D0%BB%D0%BD%D0%B0%20%D0%A1%D1%82%D1%80%D0%B0%D1%82%D0%B5%D0%B3%D0%B8%D1%98%D0%B0%20%D0%B7%D0%B0%20%D0%98%D0%9A%D0%A2%202021-2025%20%D0%B21.1.docx

Government of the Republic of Serbia. (2014). *LAW ON ELECTRONIC COMMUNICATIONS*. https://www.ratel.rs/storage/upload/2025/02/Law-on-Electronic-Communications-rev-June-2014.pdf

Government of the Republic of Serbia. (2019). *THE STRATEGY FOR THE DEVELOPMENT OF ARTIFICIAL INTELLIGENCE IN THE REPUBLIC OF SERBIA FOR THE PERIOD 2025-2030* (No.30/18). https://www.srbija.gov.rs/tekst/en/149169/strategy-for-the-development-of-artificial-intelligence-in-the-republic-of-serbia.php

Hallinan, D., & Friedewald, M. (2012). Public Perception of Modern Surveillance technologies: A selected survey analysis of the public perception and acceptance of new surveillance technologies. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.2376651

Hogan, L., & Lasek-Markey, M. (2024). Towards a human Rights-Based approach to ethical AI governance in Europe. *Philosophies*, *9*(6), 181. https://doi.org/10.3390/philosophies9060181

Hoxhaj, A. (2020). The EU rule of law initiative towards the Western Balkans. *Hague Journal on the Rule of Law*, *13*(1), 143-172. https://doi.org/10.1007/s40803-020-00148-w

Human Rights Watch. (2023, July 12). EU: Artificial intelligence regulation should protect people's rights. https://www.hrw.org/news/2023/07/12/eu-artificial-intelligence-regulation-should-protect-peoples-rights

Institute for Technology and Society. (2024). Effects of artificial intelligence on human rights: Kosovo case. Western Balkans Fund. https://westernbalkansfund.org/wp-content/uploads/2024/12/ITS-The-Effects-of-Artificial-Intelligence-on-Human-Rights-Final-EN.pdf

Institute for Technology and Society. (2024). Impact of artificial intelligence on human rights in Kosovo. https://institutets.com/impact-of-artificial-intelligence-on-human-rights-in-kosovo

KOHA.net. (2024, April 5). Privacy Agency for cameras in Pristina: Biometric characteristics will not be recorded. *KOHA.net*. https://www.koha.net/en/arberi/agjencia-per-privatesi-per-kamerat-ne-prishtine-sdo-te-regjistrohen-karakteristikat-biometrike

Korac. (2021). *Data (De)colonization: Case of the "Safe City" Project in Belgrade* [Master of Science Programme in Development Studies, Lund University]. https://lup.lub.lu.se/luur/download?func=downloadFile&recordOId=9074826&fileOId=9074827

Krivokapić, Đ., Živković, I., & Nikolic, A. (2023). Artificial intelligence regulation in the areas of data protection, information security, and anti-discrimination in Western Balkan economies. The 45th International ICT Convention MIPRO, 2022, 1403-1409. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4563901

Multi-dimensional review of the Western Balkans. (2021). In *OECD development pathways*. https://doi.org/10.1787/4d5cbc2a-en

National Assembly of the Republic of Serbia. (2019). *Law on Health Care* (No.25/2019). https://www.jpm.law/wp-content/uploads/2019/04/New-Law-on-Healthcare.pdf

Open Society Foundations Western Balkans. (2024). DSA, DMA, AIA and Western Balkans. https://osfwb.org/wp-content/uploads/2024/07/DSA-DMA-AI-Study.pdf

Poulou, A. (2014). EU Social Rights Accession Conditionality and the Western Balkans: the Example of Bosnia and Herzegovina. *Comparative Southeast European Studies*, *62*(3), 305325. https://doi.org/10.1515/soeu-2014-620303

Regan, P. (2012). Hans-Georg Gadamer's Philosophical Hermeneutics: Concepts of Reading, Understanding and Interpretation. *META: RESEARCH IN HERMENEUTICS, PHENOMENOLOGY, AND PRACTICAL PHILOSOPHY*, *IV*(2), 286-303. https://www.metajournal.org

Ružić, N. (2021). Nationalising the general data protection regulation in Western Balkan. In *Collection Regional Law Review* (pp. 313-327). https://doi.org/10.18485/iup_rlrc.2021.2.ch19

Schoenherr. (2024, September 23). AI regulation and development in Serbia. https://www.schoenherr.eu/content/ai-regulation-and-development-in-serbia

SHARE Foundation. (2024). Digital ecosystem of the Western Balkans: From regulatory gap to systemic approach. https://sharefoundation.info/en/digital-ecosystem-of-the-western-balkans-from-regulatory-gap-to-systemic-approach/

Shtupi, I., & Prifti, A. (2025). Legal Responsibility of the Data Privacy Officer Following the New Regulation of European Union approximation in Albania: A Hermeneutical and Educational Research for Civil Society. In *Atlantis highlights in social sciences, education and humanities/Atlantis Highlights in Social Sciences, Education and Humanities* (pp. 842-855). https://www.atlantis-press.com/proceedings/iclel-24/126010166

Školník, M., & Haman, M. (2024). Drawing the line: public attitudes towards warranted and unwarranted government surveillance in European countries. *Security Journal*, *37*(4), 1447-1462. https://doi.org/10.1057/s41284-024-00426-4

Sollaku, O. (2023, October 15). AI for good governance and cybersecurity in the Western Balkans. DCAF. https://www.dcaf.ch/sites/default/files/imce/ECA/OrnelaSollaku_Young-faces2023.pdf

Spalević, Ž., & Vićentijević, K. (2022). GDPR and challenges of personal data protection. *The European Journal of Applied Economics*, *19*(1), 55-65. https://doi.org/10.5937/ejae19-36596

Stelmach, J. (2023). Hermeneutical legal theory. In *Springer eBooks* (pp. 1286-1292). https://doi.org/10.1007/978-94-007-6519-1_56

Tikos, A., & Krasznay, C. (2022). Cybersecurity in the V4 Countries-A cross-border case study. *Central and Eastern European eDem and eGov Days*, *335*, 163-174. https://doi.org/10.24989/ocg.v335.13

United Nations. (2021). *Recommendation on the Ethics of Artificial Intelligence*. United Nations Educational, Scientific and Cultural Organization. https://unesdoc.unesco.org/ark:/48223/pf0000381137

Zejnullahu, N. (2020). Kosovo · Personal data protection in Kosovo, three years of failure. *European Data Protection Law Review*, *6*(2), 308-313. https://doi.org/10.21552/edpl/2020/2/18